

SYNOPSIS ON
“WEATHER REPORTING OVER IOT”

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of
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In
Computer Science and Engineering
(Internet of Things)
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Abstract

The rapid growth of Internet of Things (IoT) technology has made it possible to monitor environmental conditions in real time with the help of smart devices and cloud-based systems. This project presents an IoT-based weather reporting system using the ESP32 microcontroller, DHT sensor, rain sensor, and Supabase cloud database. The main objective of the project is to measure important weather parameters such as temperature, humidity, and rainfall, and make this information available to the user instantly through a computer dashboard.

In this system, the DHT sensor is used to measure temperature and humidity, while the rain sensor detects the presence of rainfall. These sensors are connected to the ESP32 through a breadboard and jumper wires. The ESP32 continuously reads the sensor values and processes them with the help of a program written in the Arduino IDE. Since the ESP32 has built-in Wi-Fi capability, it can directly connect to the internet without requiring any additional communication module.

After collecting the sensor readings, the ESP32 sends the data to Supabase through a wireless internet connection. Supabase acts as a cloud database where all the readings are stored securely in real time. The stored data can then be displayed on a computer screen or web dashboard. This allows users to monitor the current weather conditions from any location. The cloud storage feature also makes it possible to save previous records and observe changes in weather over time. The project was tested under different environmental conditions, and the system produced accurate and reliable results. Temperature and humidity values changed correctly according to the surroundings, and the rain sensor responded immediately when water was detected. The communication between the ESP32 and Supabase remained stable with very little delay. The project therefore demonstrates that IoT technology can be effectively used for weather monitoring and real-time data collection.

Overall, the proposed system is low-cost, simple to implement, and suitable for both educational and practical applications. It can be used in homes, schools, farms, and small weather stations where continuous environmental monitoring is required. In the future, the project can be improved by adding more sensors, mobile application support, alert notifications, and advanced data analytics to make the system more efficient and user-friendly.

Introduction

In recent years, the rapid advancement of Internet of Things (IoT) technologies has enabled the development of smart systems capable of monitoring and controlling real world environments in real time. Weather monitoring is a critical application area that impacts sectors such as agriculture, environmental management, disaster prediction, and smart city development. Traditional weather monitoring systems are often expensive, bulky, and limited to specific locations, making them less accessible for small scale or individual use. Therefore, there is a growing need for low cost, efficient, and scalable solutions that can provide accurate environmental data remotely.

The proposed project, IoT-Based Weather Monitoring System using ESP32 and Cloud Integration, aims to address this need by designing a smart system capable of measuring parameters such as temperature, humidity, and rainfall status. The system utilizes the ESP32 microcontroller, which has built in Wi-Fi capabilities, allowing seamless communication with cloud platforms. Sensors continuously collect environmental data, which is processed and transmitted in real time using HTTP protocols in JSON format. This ensures efficient data handling and enables users to monitor environmental conditions from anywhere through internet connected devices. Additionally, the system is designed to operate with minimal power consumption and high reliability, making it suitable for continuous monitoring applications. The use of lightweight communication protocols further improves the speed and efficiency of data transmission

Furthermore, the integration of cloud platforms such as Supabase enhances the system by enabling real time data storage, retrieval, and analysis. The collected data is organized in a structured database, allowing users to visualize trends and access historical records. This project not only demonstrates the practical implementation of IoT concepts but also highlights the potential of combining embedded systems with cloud computing to create intelligent and cost effective monitoring solutions. Overall, the system provides a reliable, scalable, and user friendly approach to modern weather monitoring.

Project Objective

The primary objective of this project is to design and develop a smart and efficient IoT based weather monitoring system using the ESP32 microcontroller and cloud integration. The system focuses on real time data collection, transmission, and storage of environmental parameters to enable remote monitoring and analysis.

- **Data Acquisition:** The system aims to accurately measure environmental parameters such as temperature, humidity, and rainfall using appropriate sensors. These sensors continuously collect real time data from the surroundings and provide input to the ESP32 microcontroller for processing.
- **Real Time Data Transmission:** Another key objective is to enable seamless transmission of collected data to a cloud platform using Wi-Fi connectivity. The ESP32 sends data in JSON format through HTTP POST requests, ensuring efficient and structured communication between the device and the server.
- **Cloud Storage and Integration:** The system integrates with a cloud platform (Supabase) to store sensor data in real time. Data is sent using HTTP requests in structured JSON format. The cloud database organizes data into tables for easy access and management. This enables users to retrieve and analyze historical records efficiently. It also ensures scalability and secure remote data handling.
- **Remote Monitoring and Accessibility:** The system is designed to allow users to monitor weather conditions from anywhere using internet enabled devices. This enhances convenience and ensures that real-time updates are always available without the need for physical presence.
- **Cost-Effective and Scalable Solution:** The project aims to provide an affordable alternative to traditional weather monitoring systems by using low cost components and open source technologies. The system is also scalable, allowing future enhancements such as additional sensors or advanced analytics.
- **Reliability and Efficiency:** The system is designed to ensure consistent and accurate data collection with minimal errors. It operates continuously with stable Wi-Fi connectivity for real time updates. Efficient data transmission using lightweight protocols improves performance. This ensures reliable monitoring and better system responsiveness.

Literature Review:

The field of weather monitoring has seen significant advancements with the introduction of IoT and embedded systems. Earlier systems were primarily manual or semi-automated, requiring human intervention for data collection and analysis. These traditional methods lacked real-time accessibility and were often limited in scope and scalability. With the evolution of microcontrollers and wireless communication technologies, modern systems have shifted towards automated data acquisition and remote monitoring. Studies and existing implementations have demonstrated the effectiveness of IoT in collecting environmental data and transmitting it over the internet for analysis and visualization.

Recent research has focused on developing smart weather monitoring systems using platforms such as Arduino, Raspberry Pi, and ESP based microcontrollers. Among these, ESP32 has gained popularity due to its built in Wi-Fi capabilities, low power consumption, and cost effectiveness. Many systems integrate sensors for temperature, humidity, and rainfall detection, and use cloud platforms to store and visualize data. Additionally, the use of lightweight communication protocols such as HTTP and JSON has improved the efficiency of data transmission. However, some existing solutions are either complex in design or depend on expensive infrastructure, making them less suitable for small scale or educational use.

Furthermore, cloud based IoT platforms such as Firebase, AWS IoT, and Supabase have enhanced the functionality of monitoring systems by enabling real-time data storage, remote accessibility, and analytics. These platforms allow users to track environmental changes over time and make data driven decisions. Despite these advancements, challenges such as data security, system reliability, and seamless integration still exist. The proposed project builds upon these existing studies by offering a simplified, cost effective, and efficient solution using ESP32 and Supabase, making it suitable for real world applications as well as academic implementation. Additionally, the system ensures structured data handling and easy visualization through cloud databases. It also provides flexibility for future enhancements such as advanced analytics and alert systems. This makes the solution more practical and adaptable to evolving technological needs.

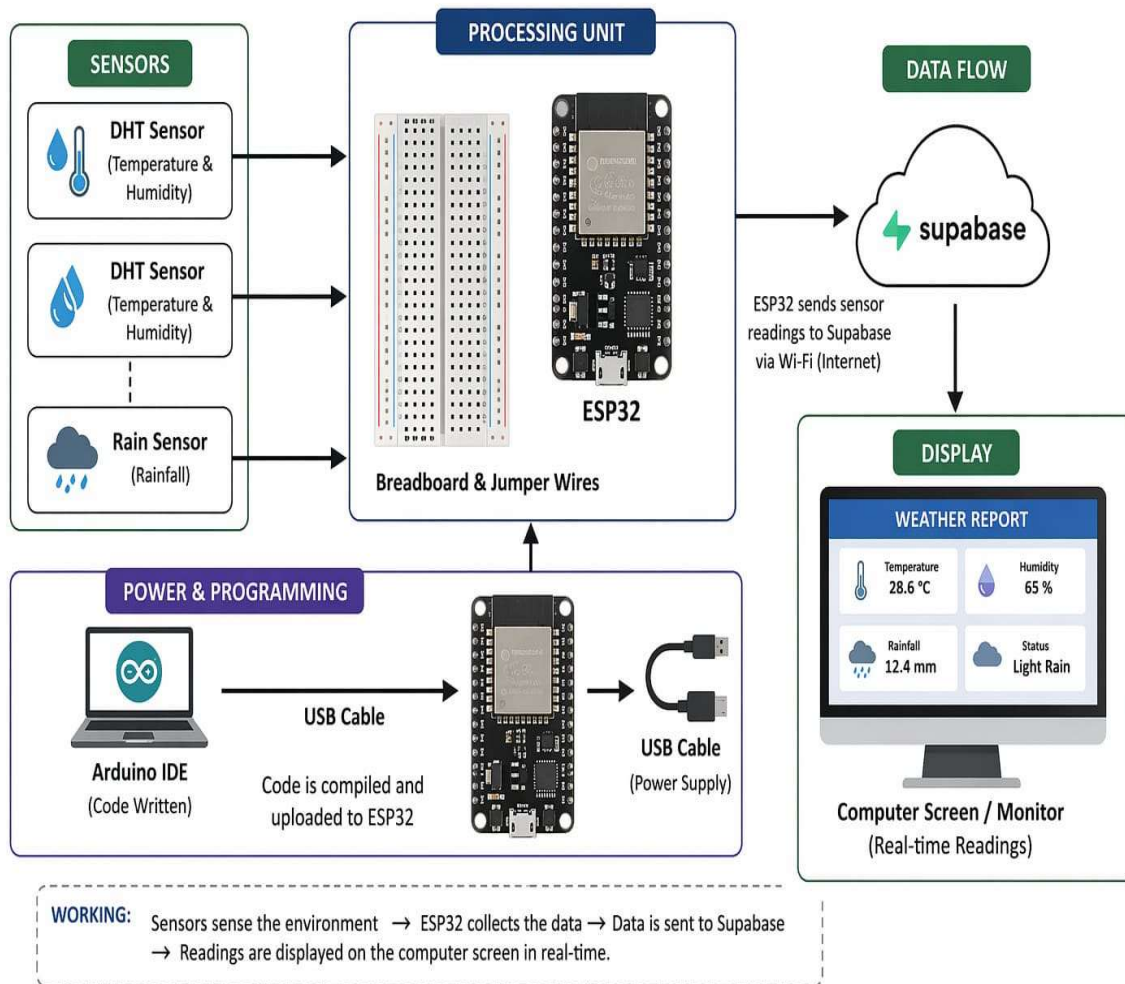
Components Required

The following components are required for the implementation of the IoT based weather monitoring system:

- **ESP32 Dev Module** – Serves as the main microcontroller with built-in Wi-Fi for data communication.
- **Temperature and Humidity Sensor (DHT11)** – Measures environmental temperature and humidity.
- **Jumper Wires** – Used to establish electrical connections between components.
- **Breadboard / PCB** – Used for circuit assembly and prototyping.
- **Data cable (USB)** – Provides necessary power to the ESP32 and sensors. Responsible for data transfer
- **Rain sensor** : detects rainfall, returns True or False value based on the rainfall occurrence.

Block Diagram

IOT BASED WEATHER REPORTING SYSTEM OVER ESP32



The block diagram shows the complete working of the IoT-based weather reporting system. The DHT sensor and rain sensor collect environmental data such as temperature, humidity, and rainfall and send it to the ESP32 through the breadboard and jumper wires. The ESP32 processes this data and, using its built-in Wi-Fi, uploads it to Supabase. The data is then stored, analyzed, and displayed on the computer screen or dashboard in real time. The Arduino IDE is used to write and upload the code to the ESP32 through a USB cable.

Tools/Technology Used:

1. Hardware Requirements

The hardware components used are simple, cost effective, and easily available:

- **ESP32 Dev Module** – Acts as the main microcontroller with built-in Wi-Fi for IoT communication.
- **Temperature and Humidity Sensor (DHT11)** – Measures environmental temperature and humidity.
- **Rain Sensor** – Detects rainfall or presence of water.
- **Breadboard** – Used for assembling the circuit without soldering.
- **Jumper Wires** – Used for connections between components.
- **Power Supply (5V/USB)** – Provides power to ESP32 and sensors.
- **Data Cable (USB)** – for data transfer and uploading code to esp-32

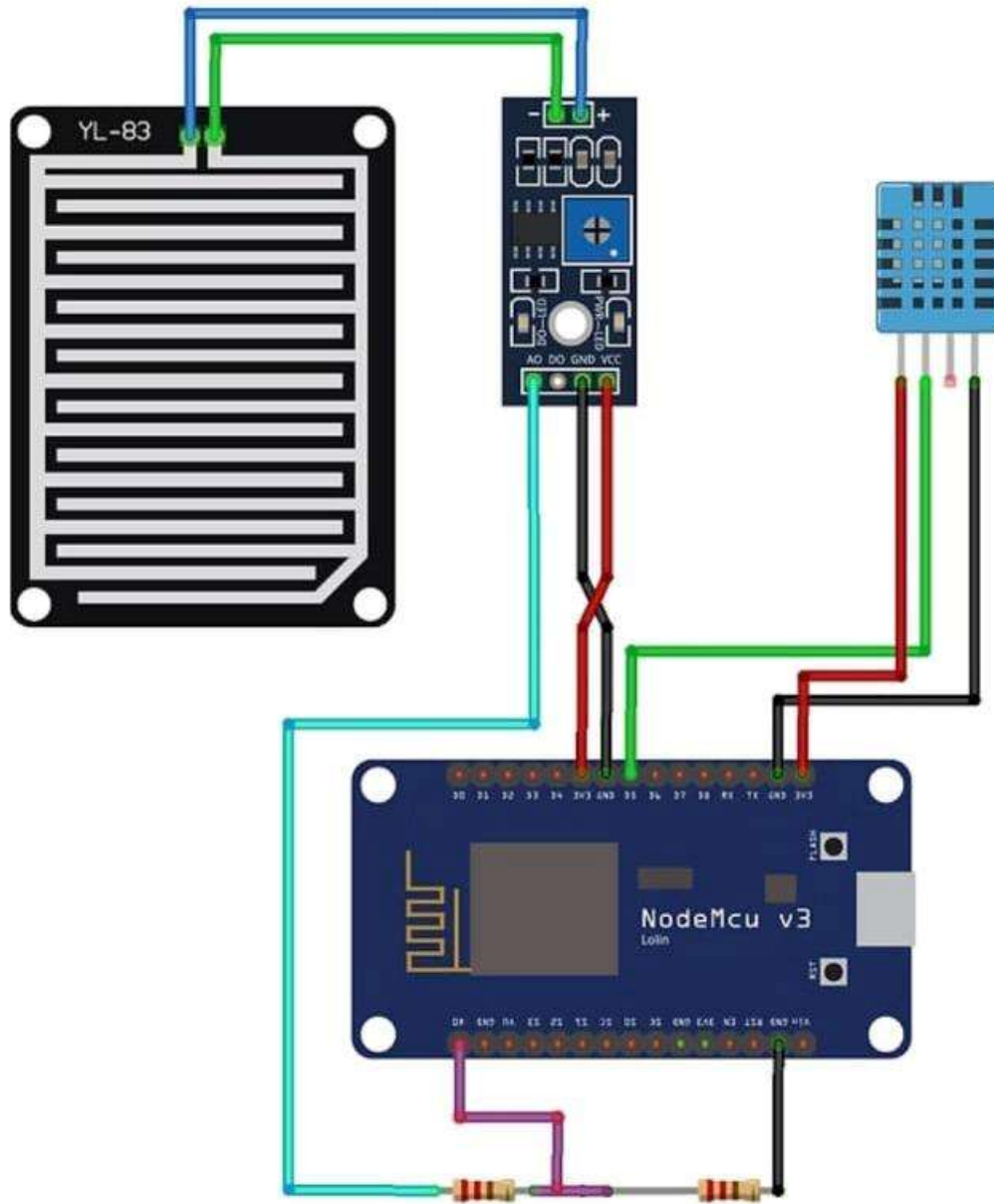
2. Software Requirements

The software tools used for development and implementation are as follows:

- **Arduino IDE** – Used for coding, compiling, and uploading programs to ESP32.
- **Programming Language (C/C++)** – Used to write the logic for sensor reading and data transmission.
- **Libraries used** – include WiFi for connecting ESP32 to the internet, HTTPClient for sending HTTP requests, and ArduinoJson for creating and handling JSON data, DHT sensor library.
- **Cloud Platform (Supabase)** – Used for storing and managing sensor data in real time.
- **REST API** – Used for communication between ESP32 and cloud database.

Circuit Diagram

Circuit Diagram

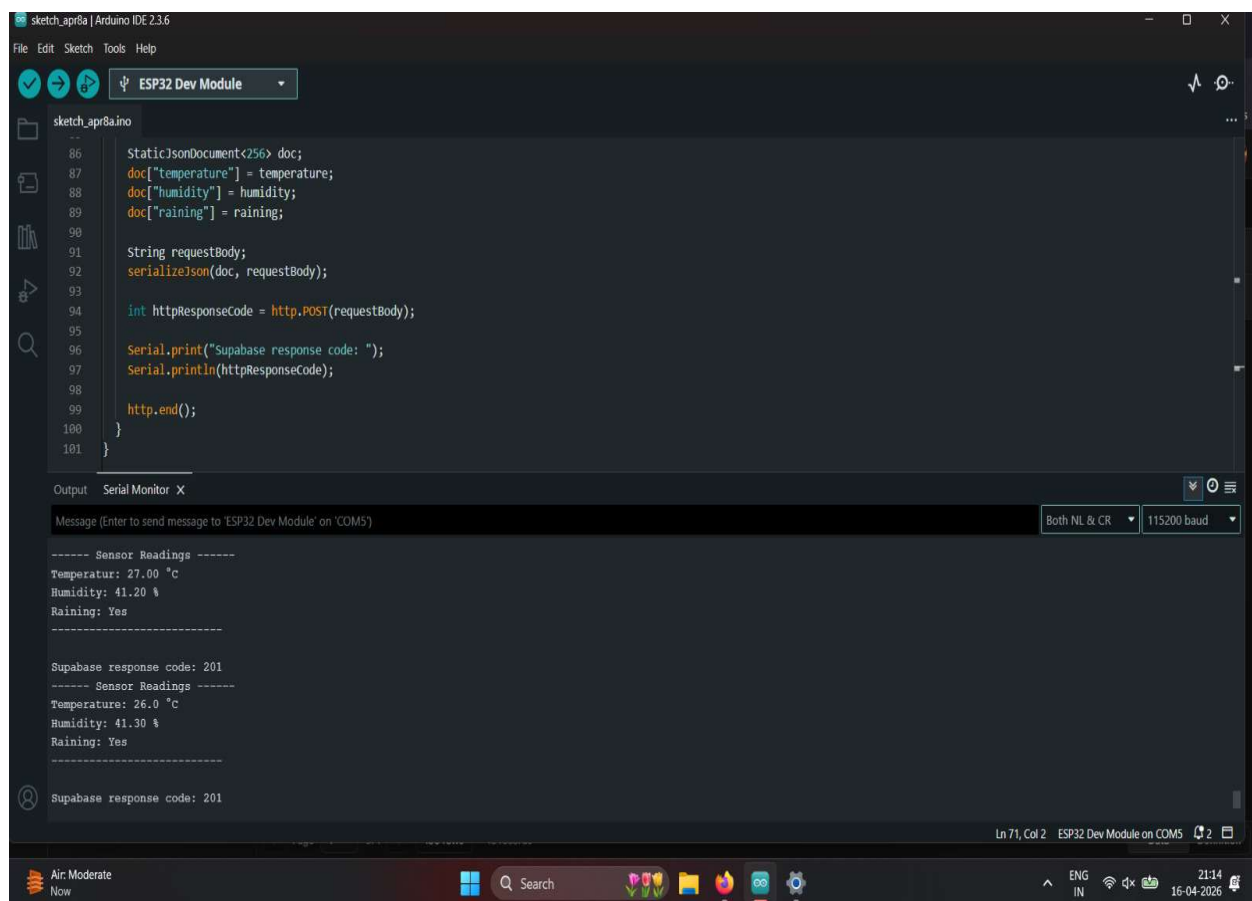


Software Implementation

Platform: The software implementation is carried out using Arduino IDE, where the ESP32 is programmed to read data from sensors such as temperature, humidity, and rain sensors.

Libraries: WiFi, HttpClient, ArduinoJson DHT sensor library are used for network communication and data handling.

Working: The software implementation of the project begins with writing the program in the Arduino IDE and uploading it to the ESP32 through a USB cable. The program continuously reads temperature and humidity values from the DHT sensor and rainfall data from the rain sensor. These readings are processed by the ESP32 and converted into suitable values for transmission. Using the built-in Wi-Fi module, the ESP32 connects to the internet and sends the sensor data to Supabase, where it is stored in real time. The stored data is then fetched and displayed on the computer dashboard, allowing the user to monitor current weather conditions and view updated readings automatically.



```
sketch_apr8a.ino | Arduino IDE 2.3.6
File Edit Sketch Tools Help
ESP32 Dev Module

sketch_apr8a.ino
86 StaticJsonDocument<256> doc;
87 doc["temperature"] = temperature;
88 doc["humidity"] = humidity;
89 doc["raining"] = raining;
90
91 String requestBody;
92 serializeJson(doc, requestBody);
93
94 int httpResponseCode = http.POST(requestBody);
95
96 Serial.print("Supabase response code: ");
97 Serial.println(httpResponseCode);
98
99 http.end();
100 }
101 }
```

Output Serial Monitor X

Message (Enter to send message to 'ESP32 Dev Module' on 'COM5')

Both NL & CR 115200 baud

```
----- Sensor Readings -----
Temperature: 27.00 °C
Humidity: 41.20 %
Raining: Yes
-----

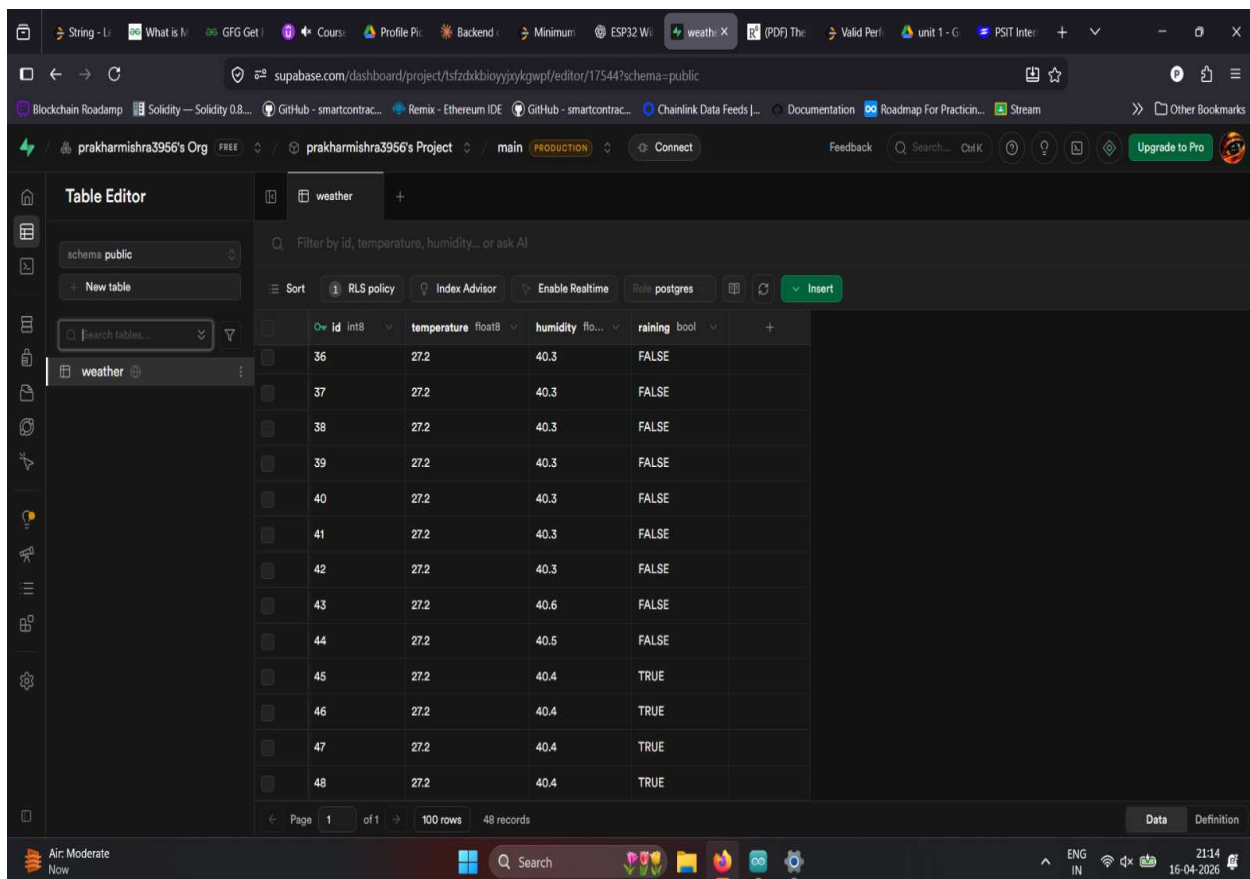
Supabase response code: 201
----- Sensor Readings -----
Temperature: 26.0 °C
Humidity: 41.30 %
Raining: Yes
-----

Supabase response code: 201
```

Ln 71, Col 2 ESP32 Dev Module on COM5

Cloud and App Integration

The software implementation is carried out using Arduino IDE, where the ESP32 is programmed to read data from sensors such as temperature, humidity, and rain sensors. Libraries like WiFi, HTTPClient, and ArduinoJson are used for network communication and data handling. The system converts sensor readings into JSON format and sends them to the cloud using HTTP POST requests. The serial monitor is used to verify readings and response codes, ensuring proper functioning of the system.



The screenshot displays the Supabase Table Editor interface for a project named 'prakharmishra3956's Project'. The table being edited is named 'weather' and is located in the 'public' schema. The table has four columns: 'id' (integer), 'temperature' (float), 'humidity' (float), and 'raining' (boolean). The table contains 48 records. The interface includes a sidebar with navigation icons, a top bar with project information, and a bottom status bar.

id	temperature	humidity	raining
36	27.2	40.3	FALSE
37	27.2	40.3	FALSE
38	27.2	40.3	FALSE
39	27.2	40.3	FALSE
40	27.2	40.3	FALSE
41	27.2	40.3	FALSE
42	27.2	40.3	FALSE
43	27.2	40.6	FALSE
44	27.2	40.5	FALSE
45	27.2	40.4	TRUE
46	27.2	40.4	TRUE
47	27.2	40.4	TRUE
48	27.2	40.4	TRUE

Data Analytics

Date	Time	Temp(°C)	Humidity(%)	Rain status
20-04-2026	10:05 AM	27.0	41.2	Yes
20-04-2026	01:12 PM	26.0	41.3	Yes
21-04-2026	09:44 AM	27.2	40.3	No

Data collected from the sensors is stored in the Supabase cloud database. This allows real time monitoring and analysis of environmental conditions. The stored data can be used to observe patterns in temperature, humidity, and rainfall over time. It also helps in understanding weather trends and making informed decisions based on historical data.

 id int8	temperature float8	humidity flo...	raining bool	+
36	27.2	40.3	FALSE	
37	27.2	40.3	FALSE	
38	27.2	40.3	FALSE	
39	27.2	40.3	FALSE	
40	27.2	40.3	FALSE	
41	27.2	40.3	FALSE	
42	27.2	40.3	FALSE	
43	27.2	40.6	FALSE	
44	27.2	40.5	FALSE	
45	27.2	40.4	TRUE	
46	27.2	40.4	TRUE	
47	27.2	40.4	TRUE	
48	27.2	40.4	TRUE	

Page 1 of 1 → 100 rows 48 records

Results and Discussion

The IoT-based weather reporting system successfully collected environmental data using the DHT sensor and rain sensor connected to the ESP32. The DHT sensor measured temperature and humidity, while the rain sensor detected rainfall conditions. These sensor values were continuously read by the ESP32 and prepared for transmission.

The ESP32 transmitted the collected readings to Supabase through a Wi-Fi connection. During testing, the communication between the ESP32 and Supabase was smooth and reliable, with only a very small delay in updating the readings. The system remained stable during continuous operation and did not show major connection failures.

The sensor readings were generally accurate and changed according to the surrounding environment. Temperature values increased when the sensor was exposed to heat, humidity values changed with moisture in the air, and the rain sensor responded immediately when water droplets were placed on it. This showed that the sensors were functioning properly and giving realistic results.

Supabase successfully stored all the collected sensor data in real time. Because the readings were saved in the cloud database, it became possible to observe both current and previous weather conditions. The stored data can also be used for basic data analysis, such as identifying temperature changes, humidity variations, and rainfall patterns over time.

The system performed efficiently and consumed very little power, making it suitable for long-term use. The dashboard on the computer screen displayed the data clearly and updated automatically whenever new values were received. Although the project depends on a stable internet connection and minor sensor errors may occur, the overall performance was satisfactory and the project successfully achieved its goal of real-time weather monitoring.

Another important observation during testing was that the system could store and retrieve data continuously without affecting its performance. Even when multiple readings were uploaded over a long period of time, Supabase handled the data efficiently and no records were lost. This indicates that the project can be scaled for larger applications where weather data needs to be monitored for several days or weeks. The successful integration of cloud storage and sensor monitoring shows that the system is suitable for future expansion and practical use.

Conclusion and Future Work

The developed IoT-based weather reporting system proved to be an effective and reliable solution for monitoring environmental conditions in real time. By using the ESP32 along with the DHT sensor and rain sensor, the system was able to collect temperature, humidity, and rainfall data accurately. The collected data was successfully transmitted to Supabase through Wi-Fi and displayed on the computer dashboard without noticeable delay.

One of the major advantages of the project is its low cost and simplicity. The ESP32 provides built-in Wi-Fi capability, which reduces the need for additional communication modules. The use of Supabase also made the system more efficient because it allowed the sensor readings to be stored securely in the cloud and accessed at any time. This makes the project suitable for homes, schools, farms, and small weather monitoring stations.

The project also demonstrated the usefulness of cloud-based data storage and basic data analytics. Since the readings are stored continuously, users can compare current weather conditions with previous records and observe changes over time. The system therefore provides not only real-time monitoring but also the possibility of identifying weather trends and patterns.

Although the system performed successfully, there are several areas where it can be improved in the future. More sensors can be added to measure other environmental parameters such as air pressure, wind speed, light intensity, and air quality. A more advanced dashboard can also be developed to display graphs, charts, and historical reports in a more attractive and user-friendly manner.

In future work, the system can be expanded by adding alert and notification features. For example, the system could automatically send a message or email when the temperature becomes too high or when heavy rainfall is detected. Mobile application support and predictive analysis using machine learning can also be included to make the system smarter and more useful. These improvements would make the project more advanced and suitable for large-scale weather monitoring applications.

References [IEEE format]:

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- Reference article: <https://www.dofbot.com/post/web-server-based-weather-monitor-using-dht-11-rain-sensor>
- Blynk Integration Guides, Available: <https://docs.blynk.io>